

THE INTERNET AND AI: Multipliers Of National Progress

The next global superpowers may be defined less by oil, geography, or military strength, and more by intelligence, connectivity, and speed. In earlier industrial eras, economic strength depended heavily on physical infrastructure. The new economy increasingly depends on software, semiconductors, algorithms, artificial intelligence, and digital reach. This transition creates a historic opportunity for India. But technology waves reward speed, and once ecosystems mature elsewhere, catching up becomes significantly harder.



JANARDHANA SWAMY is an electronics engineer, inventor with patents in India and the USA, and a former Member of Parliament from Chitradurga. This series draws on his rare blend of technical depth, global experience, and lifelong passion for India's leadership.

Education builds capability. Innovation converts capability into value. The internet and AI multiply both at an unprecedented scale. This is where the pace of national progress accelerates.

For most of human history, progress moved slowly. Distance limited collaboration. Infrastructure limited access. Knowledge spread gradually. Entire generations were needed to transform industries and economies. That cycle is collapsing. The internet removed distance. AI is removing effort. Together, they are compressing decades of progress into years. They are not optional upgrades. They are force multipliers.

When years collapse into weeks

A student in a small town now has access to capabilities once limited to elite institutions. A startup with five people can build products that once required hundreds or even thousands of engineers.

Designers can prototype faster. Programmers can write and debug software more quickly. Researchers can analyse enormous datasets in minutes. What once took years can now happen in weeks. This is not an

incremental improvement. It is a change in the speed of progress itself. The countries that understand this shift early will move ahead rapidly. Those who move slowly will struggle to keep pace.

India entered previous industrial revolutions late, while manufacturing leadership, large-scale R&D ecosystems, and deep technology ownership emerged elsewhere. The internet and AI present another historic opening, and India cannot afford to miss it.

The internet became infrastructure

The internet is no longer merely a communication tool. It has become infrastructure. In the industrial age, governments built roads, dams, and power lines. In the digital age, connectivity is the new lifeline.

Commerce, education, healthcare, banking, logistics, governance, manufacturing, and entertainment now depend on digital connectivity. Once connectivity reaches a national scale, opportunity scales with it. India has already demonstrated this capability.

UPI transformed digital payments. Aadhaar enabled identity at the population scale. CoWIN coordinated vaccination delivery for billions of doses. Pradhan Mantri Bharatiya Janaushadhi Pariyojana (PMBJP), or the Prime Minister's Indian Public Medicine Scheme, is providing quality generic medicines at affordable prices across India at massive scale. India Stack created digital infrastructure that enabled large-scale innovation across sectors. Scale changes economics.

A capable individual no longer needs to live in a major city to participate in the economy. Small businesses can access national markets. Students can learn from

